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ASSIGNMENT - 1

COURSE

Deep Learning Architectures and Techniques (ETMMDL274)

SUBMITTED TO

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MCA (AI & ML) | Sem: 2nd

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Academic Session: 2025 - 2027

Introduction

This lab assignment focuses on implementing a Multi-Layer Perceptron (MLP) model for binary classification using a credit risk dataset. The objective is to predict whether a customer is classified as a good or bad credit risk based on various financial and demographic attributes. The experiment demonstrates the practical application of feedforward neural networks, activation functions, loss functions, and optimization techniques in solving structured data classification problems.

Dataset Description

The German Credit dataset is a structured tabular dataset commonly used for binary classification tasks in financial risk analysis. The dataset contains **1000 instances** and **21 attributes**, including both numerical and categorical features.

The target variable is **"credit_risk"**, where:

- **0** represents bad credit (high-risk customer)
- **1** represents good credit (low-risk customer)

The dataset includes financial and demographic attributes such as:

- Credit history
- Loan amount
- Age
- Employment duration
- Savings account status
- Housing status
- Installment rate

These features collectively help in determining whether a customer is likely to repay a loan or default. The dataset is widely used in machine learning research for evaluating classification algorithms.

Data Preprocessing

Before training the neural network model, several preprocessing steps were applied to prepare the dataset for effective learning.

1. **Handling Missing Values:**
   Any missing values in the dataset were replaced using mode imputation to ensure data completeness and prevent errors during training.
2. **Encoding Categorical Variables:**
   Since neural networks require numerical input, all categorical features were converted into numerical format using the LabelEncoder technique.
3. **Feature Scaling:**
   The numerical features were standardized using the StandardScaler method. This ensures that all features have a mean of 0 and a standard deviation of 1, allowing the neural network to converge faster and perform better.
4. **Train-Test Split:**
   The dataset was divided into training and testing sets using an 80%–20% split. The training data was used to train the model, while the testing data was used to evaluate its performance.

These preprocessing steps ensured that the dataset was clean, normalized, and suitable for training a Multi-Layer Perceptron model

Model Architecture

A Multi-Layer Perceptron (MLP) model was designed for binary classification of credit risk. The architecture consists of an input layer, two hidden layers, and one output layer.

- **Input Layer:**
  The input layer receives 20 numerical features derived from the preprocessed dataset.
- **Hidden Layer 1:**
  The first hidden layer contains 64 neurons with the ReLU (Rectified Linear Unit) activation function. ReLU introduces non-linearity and helps the model learn complex patterns in the data.
- **Hidden Layer 2:**
  The second hidden layer consists of 32 neurons, also using ReLU activation. This layer further refines the learned feature representations.
- **Output Layer:**
  The output layer contains 1 neuron with a Sigmoid activation function. Since this is a binary classification problem, the Sigmoid function outputs a probability value between 0 and 1.

The model consists of **3,457 trainable parameters**.

The model was compiled using:

- **Loss Function:** Binary Cross-Entropy (suitable for binary classification tasks)

- **Optimizer:** Adam optimizer (used for efficient gradient-based optimization)
- **Evaluation Metric:** Accuracy

This architecture allows the model to learn non-linear relationships between financial attributes and credit risk classification.

Training Results

The model achieved a training accuracy of approximately 84% and a validation accuracy of around 70%. The difference between training and validation accuracy indicates slight overfitting but acceptable generalization performance.

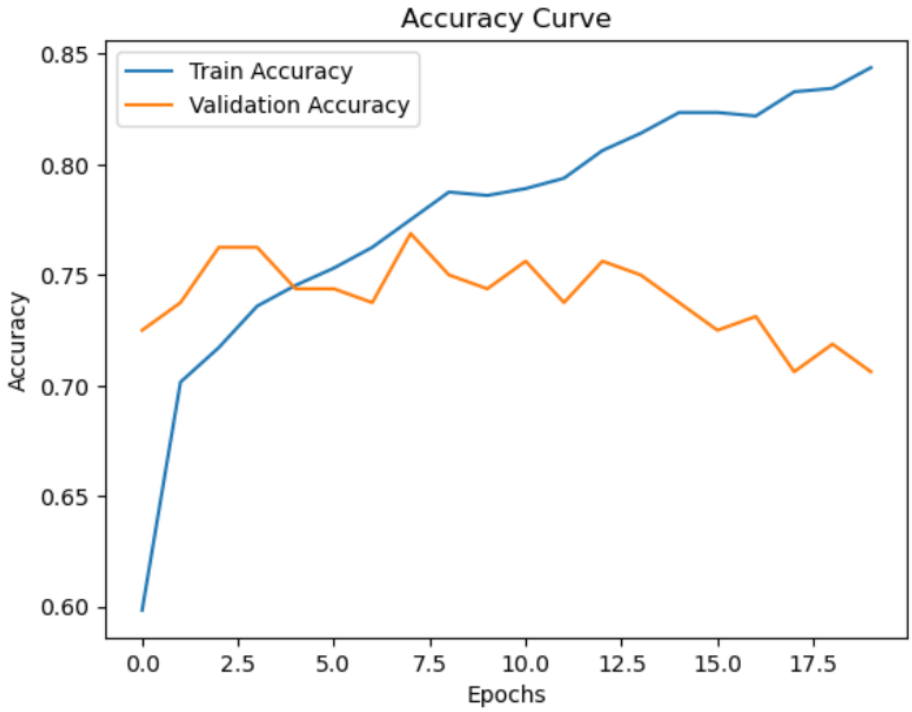

Figure 1: Training and Validation Accuracy Curve for the MLP Model

The accuracy curve shows that the model improves steadily during training. The validation accuracy stabilizes around 70%, indicating moderate generalization performance.

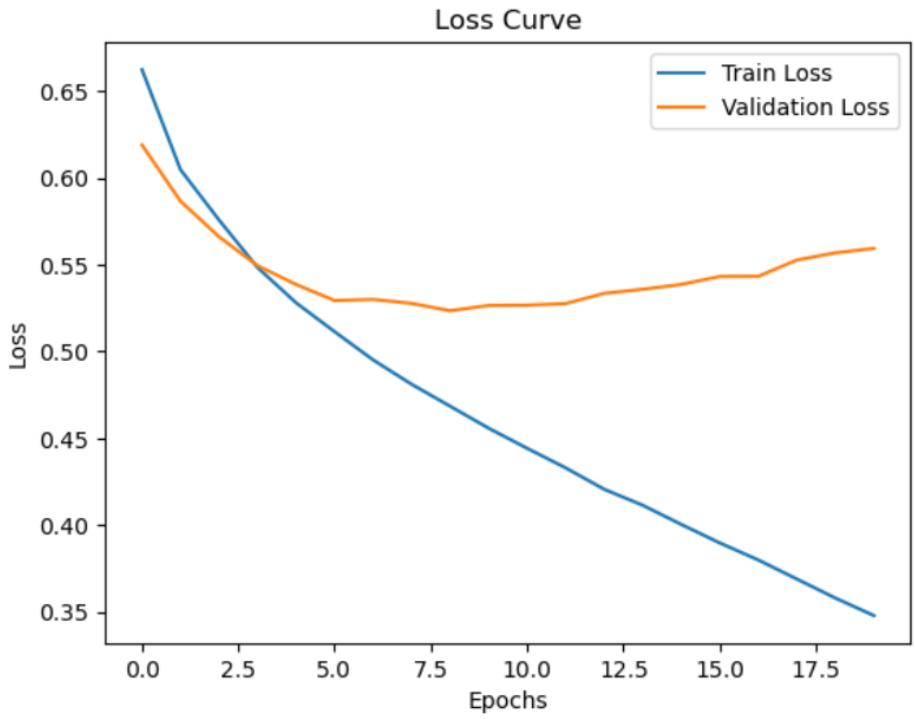

Figure 2: Training and Validation Loss Curve for the MLP Model

The loss curve demonstrates a consistent decrease in training loss. The validation loss remains relatively stable, suggesting controlled overfitting.

Evaluation Metrics

The performance of the MLP model was evaluated using a confusion matrix and classification report.

From the confusion matrix:

- True Negatives (TN): 26
- False Positives (FP): 33
- False Negatives (FN): 18
- True Positives (TP): 123

This indicates that the model correctly classified 123 good credit customers and 26 bad credit customers. However, 33 bad credit customers were incorrectly classified as good credit, which suggests some limitation in detecting high-risk cases.

Classification Report Analysis

For Class 0 (Bad Credit):

- Precision: 0.59
- Recall: 0.44
- F1-Score: 0.50

For Class 1 (Good Credit):

- Precision: 0.79
- Recall: 0.87
- F1-Score: 0.83

Overall Test Accuracy: **74%**

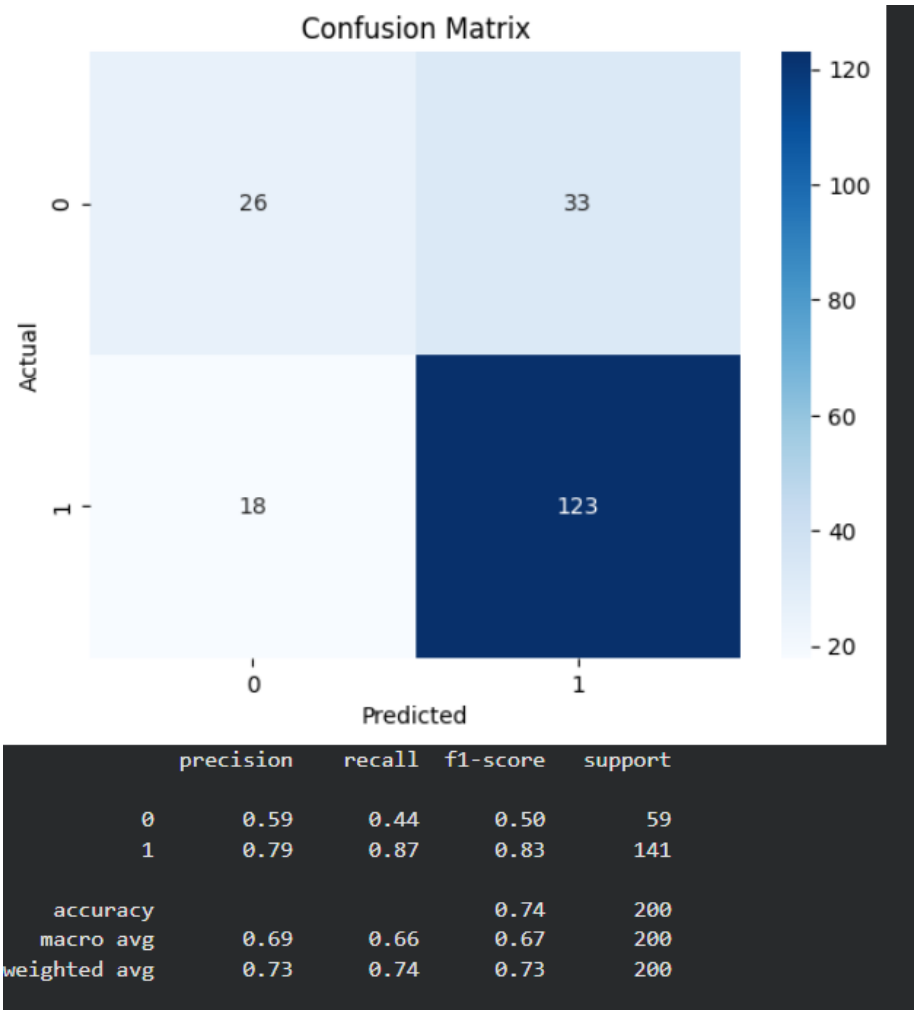

Figure 3: Confusion Matrix of the Credit Risk Classification Model

The model performs significantly better for Class 1 (good credit) compared to Class 0 (bad credit). The lower recall value for Class 0 indicates that some high-risk customers are misclassified as low-risk. This imbalance suggests that the model is biased toward predicting good credit cases more accurately than bad credit cases.

The macro average F1-score of 0.67 and weighted average F1-score of 0.73 indicate moderate overall performance.

Conclusion

In conclusion, the MLP model effectively performs binary classification for credit risk prediction with an overall test accuracy of 74%. While the model demonstrates strong performance in identifying good credit customers, it shows comparatively lower performance in detecting bad credit cases. Future improvements may include hyperparameter tuning, feature engineering, and handling class imbalance techniques.

Overall, the implementation successfully demonstrates the application of feedforward neural networks for structured binary classification tasks.